WDHN-Det: Wavelet-Coordinated Dynamic Hyperbolic Normalization for Multisource Remote Sensing Object Detection

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***Abstract*—Multisource remote sensing image object detection is essential for disaster monitoring and national defense. However, challenges remain due to heterogeneous imaging mechanisms, limited multiscale feature representation, and complex background interference. To address these issues, we propose wavelet-coordinated dynamic hyperbolic normalization for multisource remote sensing object detection (WDHN-Det), a unified object detection framework designed for robust and accurate detection in multisource remote sensing scenarios. The framework incorporates three core modules: 1) a cross-scale wavelet feature fusion (CSWFF) module that utilizes Haar wavelet decomposition in place of conventional downsampling operations, enabling cross-scale feature integration while preserving edge details and reducing high-frequency information loss; 2) a dynamic hyperbolic pyramid squeeze attention (DyHPSA) module that combines dynamic attention with hyperbolic normalization, enabling adaptive feature modulation through learnable curvature parameters to improve robustness against outliers; and 3) a cross-modal adaptive fine-grained pyramid fusion (CM-AFGPF) module that embeds dual-dimension attention into the SPPF structure, refining feature granularity and boosting detection accuracy. Extensive experiments conducted on the RSOD (optical) and SSDD [synthetic aperture radar (SAR)] datasets show that WDHN-Det achieves state-of-the-art (SOTA) performance, with mAP/precision of 95.86%/95.08% on RSOD and 98.7%/98.7% on SSDD. Compared with baseline methods, WDHN-Det improves mAP by 2.68% and 0.41% and precision by 6.68% and 0.66% on RSOD and SSDD, respectively, while maintaining high detection efficiency. These results demonstrate the effectiveness and generalization capability of WDHN-Det for multisource remote sensing object detection. The project is publicly available on https://github.com/cicadachan/WDHN-Det**

***Index Terms*—Attention mechanism, dynamic hyperbolic tangent, normalization, remote sensing, wavelet convolution.**

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I. Introduction

REMOTE sensing has become a critical technology for Earth observation, with wide-ranging applications in disaster monitoring, resource management, and urban planning. Optical remote sensing captures reflected visible and near-infrared radiation, providing high-resolution imagery that effectively reveals surface feature textures, yet is limited by cloud cover and illumination conditions. Synthetic aperture radar (SAR) actively emits microwave signals to achieve all-weather imaging capabilities and can penetrate clouds and vegetation, but speckle noise and geometric distortions result in blurred target boundaries, increasing detection difficulty.

Current multisource remote sensing object detection faces three core challenges: 1) cross-modal feature coupling difficulty—inherent differences in optical and SAR imaging mechanisms result in inconsistent feature distributions; 2) multiscale feature degradation—conventional downsampling discards high-frequency details, hindering small-object detection; and 3) complex background interference—SAR speckle noise and optical cloud occlusion increase false alarm rates. As illustrated in Fig. 1, the RSOD dataset exhibits object scale variations exceeding $10\times$ among aircraft, oil tanks, and other categories [Fig. 1(a) and (b)], while over 90% of ship targets in the SSDD dataset are smaller than 100×100 pixels and severely affected by speckle noise [Fig. 1(c) and (d)], highlighting the severity of small-object detection and noise suppression challenges.

Deeper analysis reveals that these challenges stem from inherent deficiencies in feature extraction and representation mechanisms of existing detection frameworks. First, regarding cross-modal feature coupling, traditional CNNs rely on spatially localized receptive fields for feature extraction, inadequately modeling the intrinsic relationships between optical texture details and SAR scattering intensity distributions, resulting in a significant distribution gap across feature spaces. Second, concerning multiscale feature representation, existing downsampling operations (e.g., max-pooling, average pooling, or strided convolution) reduce spatial resolution but inevitably discard high-frequency information—these components encode critical discriminative features such as target boundaries and texture details. For small objects, frequency-domain information loss directly leads to feature degradation and reduced localization accuracy. Third, regarding complex

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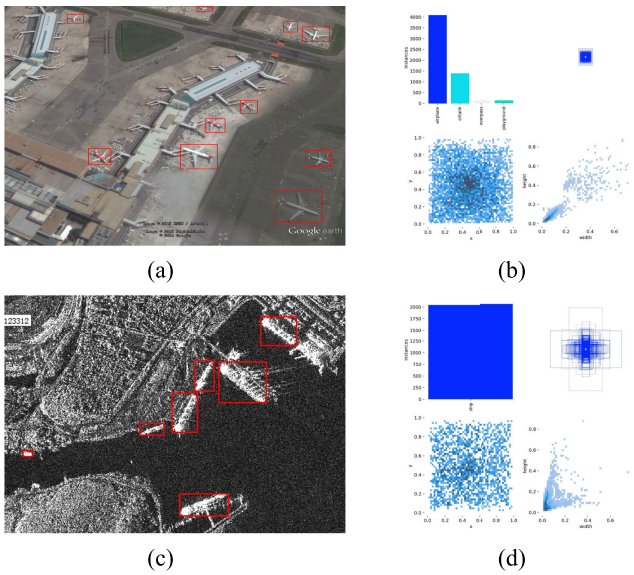


Fig. 1. Multiscale issues on the RSOD and SSDD datasets. (a) Multiscale annotation on RSOD. (b) Target distribution on RSOD. (c) Multiscale annotation on SSDD. (d) Target scale distribution on SSDD.

background interference suppression, SAR speckle noise and optical atmospheric scattering/cloud occlusion manifest as highly nonlinear amplitude fluctuations in the spatial domain. Traditional normalization methods [e.g., batch normalization (BN) and layer normalization (LN)] employ linear transformation strategies that fail to adaptively adjust feature amplitude distributions and are unable to effectively suppress gradient interference from outliers, thereby weakening model sensitivity to critical targets.

To address these issues, frequency-domain analysis and dynamic nonlinear transformations offer novel solutions. Wavelet transforms, as time–frequency joint analysis tools, decompose signals into multiscale, multidirectional frequency subbands, preserving high-frequency details while reducing resolution, thus providing a theoretical foundation for addressing multiscale feature degradation. Dynamic hyperbolic transformations, through parameterized nonlinear mapping, adaptively modulate amplitude compression strength based on input feature statistics, suppressing outliers while maintaining gradient sensitivity to targets—offering an effective mechanism for handling complex background interference. However, existing research predominantly focuses on optimizing single modalities or specific modules, lacking a unified detection framework that integrates frequency-domain feature extraction with dynamic nonlinear calibration.

For heterogeneous remote sensing data processing, current research has formed three technical paradigms. Optical image detection methods (e.g., RingMo [1] and FFCA-YOLO [2]) improve small-object detection through generative learning and feature enhancement, but their reliance on spatial-domain convolution fails to preserve high-frequency details. SAR image detection techniques (e.g., MLDet [3], HRLE-SARDet [4], SARNas [5], and SARATR-X [6]) employ multitask learning and lightweight architectures for speckle noise suppression, but their linear normalization exhibits limited capability against nonlinear noise. Cross-modal matching methods (e.g., MAP-Net [7] and MTCDN [8]) fuse heterogeneous data through deep registration and domain adaptation, yet predominantly focus on shallow feature alignment, lacking frequency-domain-level intrinsic correlation exploration and adaptive noise adjustment mechanisms. Recently, Transformer-based foundation models have demonstrated potential in single-modal remote sensing tasks, such as HyperSIGMA [9], achieving unified interpretation across tasks for hyperspectral data and MDGF-CD [10], making progress in change detection through multilevel feature fusion, but these methods have not yet effectively addressed the adaptive noise adjustment and frequency-domain deep fusion challenges for heterogeneous optical-SAR data.

Wavelet transform, as a multiresolution analysis tool, demonstrates exceptional potential in remote sensing image processing. Early research [11] achieved SAR ocean image analysis through wavelet decomposition and established theoretical frameworks, while subsequent work optimized speckle noise suppression and edge enhancement using Bayesian wavelet shrinkage [12] and multiresolution methods [13]. Recently, wavelet transform integration with deep learning has enhanced multiscale feature representation through wavelet pooling and frequency-domain feature fusion [14], [15], [16], [17], exhibiting robust spectral–spatial collaborative modeling capabilities in SAR denoising [18], [19] and cross-modal fusion [20], [21]. Wavelet convolution decomposes images into low- and high-frequency subbands via the Haar discrete wavelet transform (DWT), enabling cross-scale feature coupling while avoiding frequency-domain information loss from traditional downsampling, thus providing a theoretical foundation for resolving multiscale feature degradation.

Normalization techniques serve as key strategies for improving training stability. Since BN [22], various variants including LN [23], instance normalization (IN) [24], and group normalization (GN) [25] have been widely applied in CNNs and Transformers. However, traditional normalization suffers from performance fluctuations in small-batch training and heterogeneous data processing. The recent "normalization-free Transformer" [26] demonstrated that optimized initialization can reduce normalization dependency, inspiring our design of a dynamic hyperbolic regularization mechanism—through the parameterized hyperbolic transformation function $\tanh(\alpha \cdot x)$ introducing learnable curvature parameter α, adaptively adjusting compression strength based on input feature statistics, maintaining high gradient sensitivity for low-amplitude optical textures while strongly suppressing high-amplitude SAR outliers, providing theoretical support for heterogeneous noise processing.

Attention mechanisms effectively enhance feature selectivity in remote sensing image processing. From channel attention-based feature enhancement methods [27], [28] to adaptive attention networks [29], attention mechanisms continually advance representation precision. Inspired by these advances, this article integrates attention mechanisms [29] with spatial pyramid pooling fusion (SPPF) to design the CM-AFGPF module, optimizing cross-modal feature fusion and channel selection strategies, enhancing model adaptability to complex remote sensing scenarios.

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Therefore, this article proposes the wavelet-coordinated dynamic hyperbolic normalization for multisource remote sensing object detection (WDHN-Det) framework, achieving unified processing of optical and SAR data through three core modules. Despite fundamental differences in imaging mechanisms, both modalities share modality-invariant high-level semantic information (land-cover categories, spatial geometric structures, and target boundaries). The three core modules fully exploit this characteristic.

1) CSWFF employs a wavelet transform to uniformly decompose optical textures and SAR scattering intensities into multiscale frequency components, establishing a comparable feature representation foundation.
2) DyHPSA implements differentiated calibration for distinct modality-specific statistical characteristics (low-amplitude noise in optical imagery and high-amplitude outliers in SAR imagery) through learnable hyperbolic transformation parameters, enabling a unified normalization mechanism to accommodate heterogeneous data.
3) CM-AFGPF focuses on cross-modal shared fine-grained semantics within the pyramid feature space while suppressing modality-specific textural variations. This design enables effective detection of heterogeneous remote sensing data within a unified architecture without requiring architecture-specific adjustments for different data sources.

The main contributions of this article are as follows.

1) *Cross-Scale Wavelet Feature Fusion (CSWFF) Module:* Utilizing the Haar wavelet transform to decompose optical and SAR imagery into four subbands—LL (low-frequency approximation), LH/HL (horizontal/vertical details), and HH (diagonal details). By coupling multiscale features during spatial resolution reduction, it effectively preserves high-frequency textures and SAR structural information, mitigating the information loss and edge blurring typically caused by conventional downsampling operations.
2) *Dynamic Hyperbolic Pyramid Squeeze Attention (DyHPSA) Module:* This module integrates pyramid squeeze attention (PSA) with dynamic hyperbolic transformation by embedding a hyperbolic normalization layer with learnable curvature parameters at stage transitions. A parameterized hyperbolic compression function adaptively regulates feature magnitudes, enhancing sensitivity to key targets while effectively suppressing outlier interference.
3) *Cross-Modal Adaptive Fine-Grained Pyramid Fusion (CM-AFGPF) Module:* This module employs a dual-dimensional attention gating mechanism across channel and spatial domains, coupled with deformable pooling to adaptively calibrate multiscale features from optical and SAR modalities. Local entropy constraints are introduced to suppress SAR background noise, while channel-wise variance enhances the semantic saliency of optical targets.

The structure of this article is as follows. Section II reviews related work, Section III introduces the proposed methodology, Section IV presents the experimental results, and Section V summarizes the main contributions and discusses future research directions.

II. Related Work

This section overviews relevant research for multisource remote sensing object detection, including optical-SAR cross-modal processing, wavelet transform. We also discuss related work on normalization techniques and attention mechanisms.

A. Optical-SAR Cross-Modal Collaborative Processing Advances

Wavelet transform, as a multiresolution analysis tool, demonstrates exceptional potential in remote sensing image processing. Mallat [30] systematically established the mathematical framework of the wavelet transform, laying the theoretical foundation for multiscale feature analysis. Early studies validated its effectiveness in SAR image processing: Rodenas and Garello [11] achieved internal wave detection and wavelength estimation in SAR ocean images through continuous and DWTs. Dai et al. [12], Argenti et al. [13], and Touzi [31] optimized SAR speckle noise suppression via Bayesian wavelet shrinkage and multiresolution methods. Tello Alonso et al. [32] developed a wavelet-based edge enhancement algorithm, significantly improving coastline extraction performance.

With deep learning advancement, the wavelet transform exhibits renewed vitality in feature extraction and object detection. For feature extraction, Wang et al. [15] proposed FDNet, integrating wavelet transform into downsampling operations to enhance scene classification accuracy; Yang et al. [16] designed SFFNet based on the Haar wavelet to strengthen multiscale feature representation; Qin et al. [20] extracted robust spectral–spatial features through dual-stream wavelet patch embedding (DWPE); Zhang et al. [33] proposed ECPW-STFN, separating high-to-low-frequency components for independent training with wavelet loss functions preserving image details. For object detection, Xue et al. [14] designed SmallTrack using wavelet pooling layers (WPLs) to decompose small target features, significantly improving UAV remote sensing tracking; Pan et al. [17] combined the Haar wavelet tree structure with multihead attention, achieving dual optimization in accuracy and efficiency on Gaofen-3 SAR and WorldView-3 optical datasets; Zhang et al. [19] introduced DWT and feature-frequency transform normalization (FFTN) in image translation, effectively preserving semantic structure and enhancing fidelity.

These studies validate the Haar wavelet transform's versatility in optical and SAR image denoising, feature enhancement, and multiscale analysis, providing theoretical and practical support for the CSWFF module design in our WDHN-Det framework.

B. Wavelet Transform

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C. Normalization Techniques

Normalization techniques are important in deep learning for enhancing network training stability and performance. Ioffe and Szegedy [22] introduced BN, which normalizes layer inputs across mini-batches, effectively accelerating training and reducing internal covariate shift. Ba et al. [23] proposed LN, normalizing features within a single sample, demonstrating strong performance in recurrent neural networks and later in Transformer architectures. Ulyanov et al. [24] introduced IN, normalizing each sample independently, achieving remarkable results in style transfer tasks. Vaswani et al. [34] employed LN in the Transformer architecture, making it a standard component for self-attention mechanisms. Wu and He [25] proposed GN, which normalizes grouped channels, addressing the limitations of BN in small-batch scenarios and showing strong adaptability. While these traditional normalization methods are effective, they often introduce additional computational overhead, and fixed normalization strategies may constrain the network's expressive capacity.

As normalization significantly enhances network performance, discussions regarding the necessity of normalization in networks have gradually increased. Zhu et al. [26] explored the possibility of removing normalization from Transformer architectures in "Transformers without normalization," demonstrating that optimized initialization and residual connections can maintain training stability while preserving the network's original dynamics. This research inspired the development of the DyHPSA module proposed in this article. DyHPSA employs a dynamic hyperbolic tangent function to replace traditional normalization operations, applied after the feed-forward network (FFN) and attention operations in the pyramid squeeze attention (PSA) network module. Unlike conventional normalization, DyHPSA dynamically adapts the activation range based on input, aiming to maintain training stability and enhance feature representation without introducing the computational overhead of normalization.

III. Methodology

A. Overview

We propose wavelet-coordinated with dynamic hyperbolic normalization for remote sensing detection to achieve cross-modal feature optimization and enhance detection robustness and accuracy in multisource remote sensing object detection, as illustrated in Fig. 2. The framework comprises three core modules working synergistically. First, the CSWFF downsampling module employs the Haar wavelet transform to decompose both optical and SAR images into four-channel subbands (LL, LH/HL, and HH), replacing traditional pooling-based downsampling to extract multiscale feature representations. Second, the DyHPSA module integrates PSA mechanisms with dynamic hyperbolic transformation, embedding learnable curvature-parameterized hyperbolic normalization layers across cross-stage connections to enhance adaptive feature amplitude adjustment and optimize target gradient sensitivity. Third, the CM-AFGPF module introduces channel–spatial dual-dimensional attention mechanisms with deformable pooling to dynamically calibrate optical or SAR feature weights, leveraging local entropy constraints to suppress SAR background noise while exploiting channel variance to emphasize optical target semantics, thereby achieving fine-grained cross-modal feature fusion. Finally, the fused features are processed through detection heads for target localization and classification, effectively improving detection performance in complex scenarios.

All three core modules of WDHN-Det adopt modality-adaptive design strategies, enabling unified processing of optical and SAR data. Specifically, CSWFF applies the same frequency-domain decomposition mechanism via the Haar wavelet transform to both modalities, deconstructing RGB/grayscale signals from optical images and scattering echo signals from SAR images into low-frequency approximations (LL) and three directional high-frequency details (LH/HL/HH), providing consistent input representation for subsequent processing; DyHPSA's learnable curvature parameter α adaptively adjusts feature amplitude distributions across modalities through data-driven optimization, with ablation experiments demonstrating this parametric adaptation as critical for cross-modal generalization; CM-AFGPF's channel–spatial attention mechanism focuses on semantic salience rather than modality-specific surface textures, designed to extract modality-shared discriminative features.

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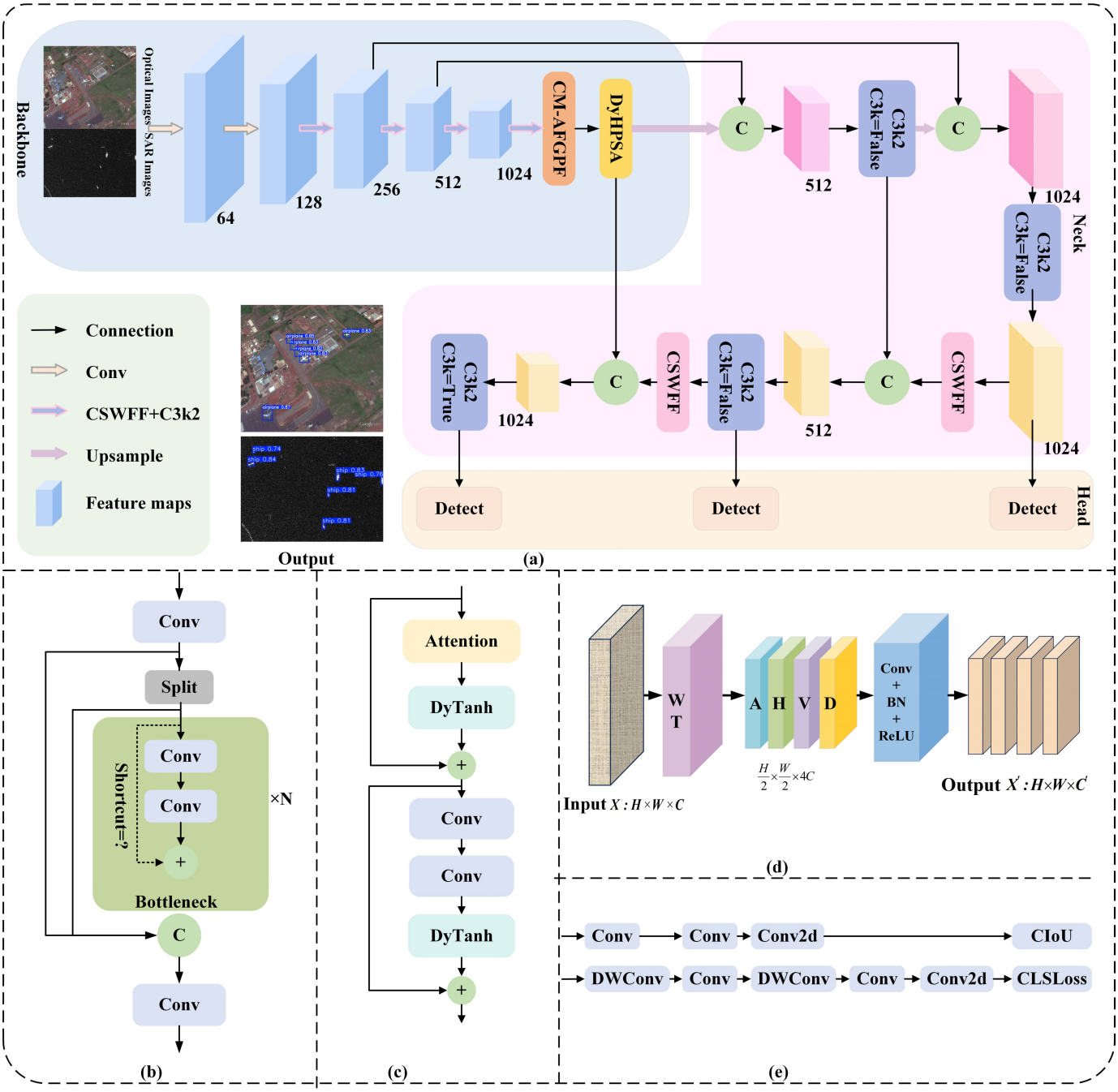


Fig. 2. WDHN-Det network architecture. The framework takes multisource remote sensing images as input, extracts multichannel features through the backbone, and enhances them collaboratively via the CM-AFGPF and DyHPSA modules, followed by the neck and detection head for detection. The C3k2 module serves as the fundamental feature extraction unit, consisting of convolution, segmentation, and residual-structured bottleneck components. DyHPSA integrates PSA with dynamic hyperbolic transformations, calibrating features through attention, DyTanh, and residual connections. CSWFF decomposes features using the wavelet transform, achieving cross-scale coupling through convolution and normalization. Finally, the detection head constructs branches using convolution and depthwise separable convolution, completing localization and classification with CIoU and CLSLoss. (a) Overall. (b) C3k2. (c) DyHPSA. (d) CSWFF. (e) Detect.

This end-to-end adaptive design eliminates the need for architecture modifications or hyperparameter retuning across different modalities.

B. CSWFF Module

The core advantages of frequency-domain transformation lie in: 1) decomposing spatial information into multiscale frequency components to decouple semantics and details; 2) preserving critical information while reducing resolution through frequency-domain subbands, thereby improving computational efficiency; and 3) providing a unified frequency-domain decomposition structure for heterogeneous remote sensing imagery (optical/SAR), facilitating subsequent cross-modal feature learning. However, different frequency-domain transformation methods exhibit significant differences in

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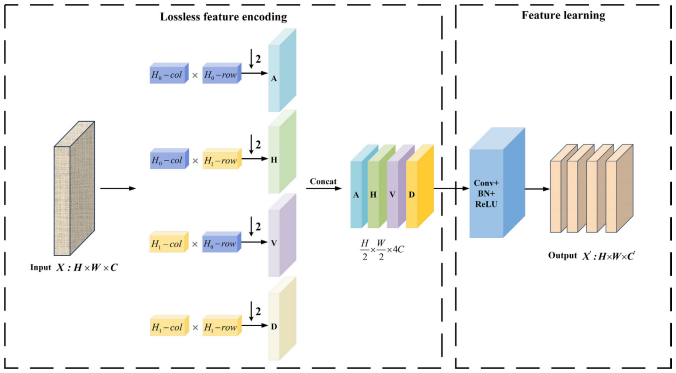


Fig. 3. Structure of the CSWFF module. The module consists of two components: lossless feature encoding and feature learning. Input features are processed by different combinations of filters to generate the low-frequency component A (approximation), horizontal edge (high-frequency) component H (horizontal), vertical edge (high-frequency) component V (vertical), and diagonal edge component D (diagonal). These components are then concatenated for feature learning, achieving efficient cross-scale feature coupling and providing highly expressive features for remote sensing target detection.

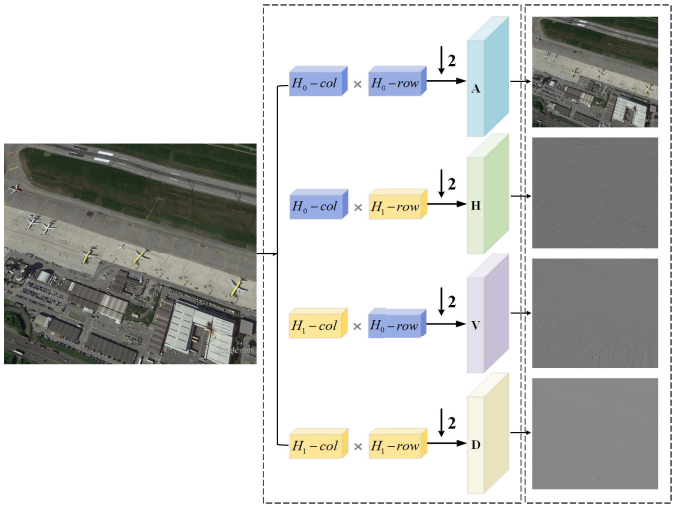


Fig. 4. Haar DWT process in the CSWFF module. This method encodes the input image into four components.

theoretical properties and practical performance, necessitating systematic comparative analysis to support design choices.

We systematically compare three mainstream frequency-domain transformation methods (Fourier transform, discrete cosine transform, and wavelet transform) in section comparative analysis of frequency-domain methods. Through energy conservation testing, frequency-domain characteristic visualization, and detection performance evaluation, experimental results demonstrate that the Haar wavelet transform more effectively extracts and preserves multiscale features in remote sensing imagery while maintaining perfect reconstruction properties. Therefore, we adopt the Haar wavelet as the core method for frequency-domain pooling.

As shown in Fig. 3, the proposed CSWFF module consists of two components: 1) a lossless feature encoding module and 2) a feature representation learning module. The lossless feature encoding module is responsible for transforming features and reducing spatial resolution, utilizing the Haar wavelet transform to efficiently downscale feature maps while fully preserving information. Subsequently, the feature representation learning module, comprising standard convolutional layers, BN layers, and ReLU activation layers, extracts discriminative features and filters redundant information.

1) Lossless Feature Encoding Module: This module leverages the Haar wavelet transform layer to effectively reduce the spatial resolution of feature maps while maintaining information integrity. The Haar wavelet transform is a widely recognized compact binary orthogonal normalized transform, extensively applied in image encoding, edge extraction, and binary logic design. In the Haar wavelet transform, the wavelet basis function and the scaling function for a one-stage 1-D Haar transform can be defined as follows:

$$\begin{cases} \phi_1(x) = \dfrac{1}{\sqrt{2}}\phi_{1,0}(x) + \dfrac{1}{\sqrt{2}}\phi_{1,1}(x) \\ \Psi_1(x) = \dfrac{1}{\sqrt{2}}\phi_{1,0}(x) - \dfrac{1}{\sqrt{2}}\phi_{1,1}(x) \end{cases} \tag{1}$$

where $\phi_{j,k}(x)$ is defined as follows:

$$\phi_{j,k}(x) = \sqrt{2^j}\phi\left(2^j x - k\right), \quad k = 0, 1, \ldots, 2^j - 1. \tag{2}$$

Here, the parameter j denotes the order of Haar basis functions or the scale in image processing, and k represents the index or the direction in 2-D images. Furthermore, $\phi_{0,0}(x)$ is defined as follows:

$$\phi_{0,0}(x) = \phi_0(x) = f(x) = \begin{cases} 1, & 0 \le x \le 1 \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

Thus, the one-stage Haar transform can be expressed using the zero-order Haar basis function as follows:

$$\begin{cases} \phi_1(x) = \phi_0(2x) + \phi_0(2x-1) \\ \Psi_1(x) = \phi_0(2x) - \phi_0(2x-1). \end{cases} \tag{4}$$

Based on this, a signal of length L can be divided into two segments of length $L/2$, corresponding to low-pass and high-pass decomposition filters, respectively. When applying the Haar wavelet transform to a 2-D signal (e.g., a grayscale image), it generates four components with half the spatial resolution of the original signal.

Fig. 4 depicts the process of dividing the image into four components using the wavelet transform. The input feature X with dimensions $H \times W \times C$ is processed through a pair of filters: the low-pass filter $H_0 = 1/\sqrt{2}[1, -1]$ and the high-pass filter $H_1 = 1/\sqrt{2}[1, 1]$. Here, H_0 – col denotes the column low-pass filter, H_1 – col denotes the column high-pass filter, H_0 – row denotes the row low-pass filter, and H_1 – row denotes the row high-pass filter. The combination of the column low-pass filter H_0 – col and row low-pass filter H_0 – row produces the approximation (low-frequency) component A (approximation). The combination of the column low-pass filter H_0 – col and row high-pass filter H_1 – row produces the horizontal edge (high-frequency) component H (Horizontal). The combination of the column high-pass filter H_1 – col and row low-pass filter H_0 – row produces the vertical edge (high-frequency) component V (vertical). The combination of the column high-pass filter H_1 – col and row

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TABLE I
COMPARISON OF PARAMETERS AND FLOPS BETWEEN SCWFF AND STRIDED CONVOLUTION

Module	Parameters	FLOPs
CSWFF	$4C^2$	$2HWC^2 + 3.75HWC$
Strided Convolution	$9C^2$	$4.5HWC^2 - 0.25HWC$

high-pass filter H_1 – row produces the diagonal edge (high-frequency) component D (diagonal). The symbol $\downarrow 2$ indicates downsampling of the approximation and detail components. Each component has a size of $H/2 \times W/2$. Notably, the resolution of each component is reduced to $H/2 \times W/2$, while the number of feature map channels increases fourfold. In other words, the Haar wavelet transform encodes part of the spatial dimension information into the channel dimension without any information loss. As a result, the image features are decomposed into multiple frequency components containing both low and high frequencies, preserving the original global information while compressing the feature maps to enhance subsequent computational efficiency.

2) Feature Representation Learning Module: The feature representation learning module consists of a standard 1×1 convolutional layer, a BN layer, and a ReLU activation function. The standard convolution is used to adjust the number of channels in the feature map. This module serves two primary functions: first, it adjusts the number of channels in the feature map to match subsequent layers; second, it filters redundant information as much as possible to enable more effective learning of representative features in subsequent layers. This module allows the CSWFF module to serve as a substitute for various downsampling methods, such as max-pooling or strided convolution.

Assuming the input feature map size is $H \times W \times C$, the feature map size after downsampling is $H/2 \times W/2 \times C'$. Compared to a traditional 3×3 convolution with a stride of 2, the total parameters and computational cost are shown in Table I. It can be observed that CSWFF performs better in terms of parameters and floating-point operations (FLOPs).

In summary, through the Haar wavelet transform, the original features are effectively decomposed into four components with halved spatial resolution, capturing low-frequency and high-frequency (horizontal, vertical, and diagonal) information from multisource remote sensing images to preserve more image details. This transformation enables the model to maintain information integrity by increasing the number of feature map channels while reducing resolution, thereby providing information-rich and resolution-appropriate inputs for subsequent layers, optimizing feature extraction, and enhancing the model's sensitivity to multisource remote sensing images.

C. DyHPSA Module

To address the heterogeneous noise distribution in optical and SAR data within multisource remote sensing imagery, this article proposes the DyHPSA module, which achieves synergistic optimization of noise suppression and multiscale feature enhancement through the deep integration of hyperbolic transformations with learnable curvature parameters and a pyramid attention mechanism. The core design revolves around the dynamic hyperbolic normalization layer, which replaces traditional linear normalization operations with a parameterized hyperbolic tangent function, mathematically defined as follows:

$$y = w \odot \tanh(\alpha \cdot x) + \beta \tag{5}$$

where α is the learnable curvature parameter (initialized as $\alpha 0 = 0.5$ for optical and SAR imagery), and w and β are the learnable weights and biases in the channel dimension.

Algorithm 1 Pseudocode of DyHPSA

```
# c: number of channels
# attn_ratio: attention ratio
# num_heads: number of attention heads
# init_alpha: initial value for DynamicTanh parameters
# shortcut: whether to use shortcut connections

class DyHPSA inherits nn.Module
def __init__(self, c, attn_ratio=0.5, num_heads=4,
             init_alpha=0.5,shortcut=True):
    super().__init__(c, attn_ratio, num_heads, shortcut)
    self.attn = Attention(c, attn_ratio=attn_ratio,
         num_heads=num_heads)
    self.ffn = nn.Sequential(
      Conv(c, c * 2, 1),
      Conv(c * 2, c, 1, act=False))
    self.dyt1 = DynamicTanh(shape=c, init_alpha=init_alpha)
    self.dyt2 = DynamicTanh(shape=c, init_alpha=init_alpha)

def forward(self, x):
    x ← x + self.attn(self.dyt1(x))
    x ← x + self.ffn(self.dyt2(x))
    return x
end class
```

The DyHPSA module constructs a synergistic optimization mechanism for dynamic feature calibration, multiscale attention focusing, and cross-layer residual learning by deeply integrating the dynamic hyperbolic normalization layer into the PSABlock architecture, effectively achieving cross-modal feature distribution alignment and enhanced noise robustness. The input feature x first undergoes modality-adaptive calibration through two independent DyT layers: dyt1 operates on the attention branch, adjusting α to enhance target region responses, while dyt2 acts on the FFN branch to suppress background noise interference. The calibrated features are then fed into an improved multihead attention module, which generates query, key, and value feature sets via convolution and splits them into $\text{num}_{\text{heads}}$ heads (default: 4), with each head's key dimension set to $\text{head}_{\text{dim}} * \text{attn}_{\text{ratio}}$ (default: 0.5) to reduce computational complexity. Attention weights are computed using a scaling factor $\text{scale} = \text{key}_{\text{dim}} - 0.5$ and Softmax normalization, combined with weighted Value features to produce a spatial-channel hybrid attention map. Additionally, depthwise separable convolution applies positional encoding to the value features, enhancing the perception of spatial

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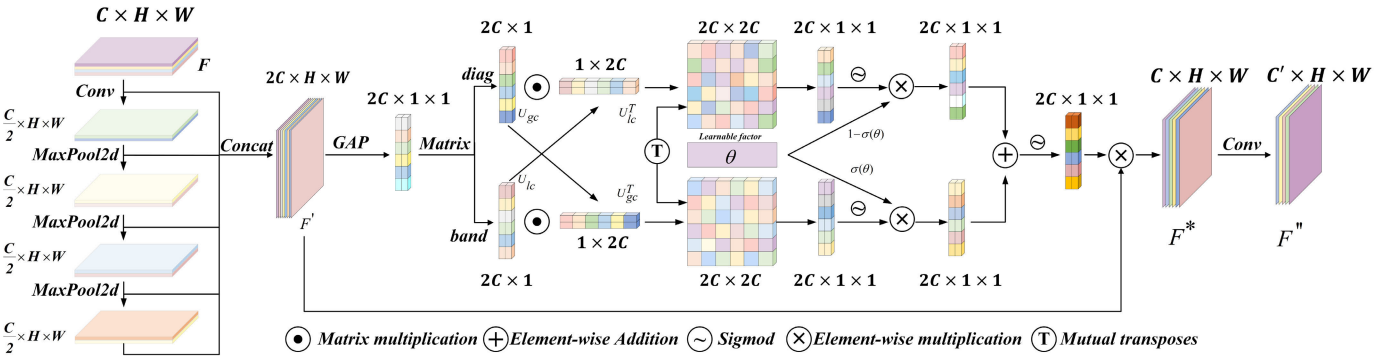


Fig. 5. Structure of the CM-AFGPF module. This module processes multiscale features through operations such as convolution, pooling, attention, and matrix transformation to achieve cross-modal feature fusion, outputting enhanced features for object detection.

distribution for ship scattering points in SAR imagery. The FFN adopts a lightweight design, achieving channel expansion ($c \to 2c$) and compression ($2c \to c$) through two convolutional layers, with the intermediate layer using the GELU activation function to enhance nonlinearity. Finally, as shown in (6), the outputs of the attention and feed-forward branches are combined through residual connections, preserving the original feature distribution and mitigating gradient vanishing issues. The pseudocode for the implementation in PyTorch is provided in Algorithm 1

$$x_{\text{out}} = x + \text{Attn}\left(\text{DyT}_1(x)\right) + \text{FFN}\left(\text{DyT}_2(x)\right). \tag{6}$$

D. CM-AFGPF Module

As shown in Fig. 5, given a feature map $F \in \mathbb{R}^{C\times H\times W}$, where C, H, and W represent the number of channels, height, and width, respectively, the feature map first undergoes a Conv operation, resulting in a feature map of size $C/2 \times H \times W$. Subsequently, three max-pooling (MaxPool2d) operations are applied, each producing a feature map of size $C/2 \times H \times W$. These feature maps are concatenated to form a feature map F' of size $2C \times H \times W$. To extract channel information from the feature map, global average pooling (GAP) is applied to convert F', which contains global spatial information, into a channel descriptor $U \in \mathbb{R}^{\mathcal{Z}^C}$. The nth channel element of U is computed as $U_n = \text{GAP}(F_n) = 1/(H \times W)\sum_{i=1}^{H}\sum_{j=1}^{W} F_n(i,j)$, where $F_n(i,j)$ is the value of the nth channel feature map at position (i,j), and GAP(x) is the GAP function. This operation transforms the shape of F from $2C \times H \times W \to 2C \times 1 \times 1$, capturing global spatial information.

Subsequently, to capture local channel information while maintaining a low number of model parameters, a band matrix $B = [b_1, b_2, b_3, \ldots, b_k]$ is used for local channel interaction, computing local information as $U_{lc} = \sum_{i=1}^{k} U \cdot b_i$ (implemented via Conv1D in experiments), where U represents the channel descriptor, U_{lc} denotes the local information, and k indicates the number of adjacent channels. Meanwhile, to obtain global channel information and enhance its representational capacity, a diagonal matrix $D = [d_1, d_2, d_3, \ldots, d_c]$ is employed to capture dependencies across all channels, deriving global information as $U_{gc} = \sum_{i=1}^{c} U \cdot d_i$ (implemented via Conv2D in experiments), where U_{gc} represents the global information and c is the number of channels.

To facilitate effective interaction between global and local information, we capture their correlations at different granularities by computing the correlation matrix $M = U_{gc} \cdot U_{lc}^T$ through a cross correlation operation between the global information U_{gc} and local information U_{lc}. Then, to accurately assign feature weights and reduce computational complexity, an adaptive fusion strategy is employed. We extract row and column information from the correlation matrix M and its transpose as weight vectors $U_{gc}^w = \sum_j^c M_{i,j}(i \in \{1,2,\ldots,\})$ and $U_{lc}^w = \sum_j^c M_{i,j}^T(i \in \{1,2,\ldots,\})$, respectively. Dynamic fusion is achieved through a learnable factorθ, defined as $W = \sigma(\sigma(\theta) \times \sigma(U_{gc}^w) + (1 - \sigma(\theta)) \times \sigma(U_{lc}^w))$, where σ is the sigmoid activation function and c is the number of channels. This approach avoids redundant cross correlation operations between local and global information, strengthens their interaction, and selectively emphasizes effective features while suppressing ineffective ones, enabling precise weight allocation for dehazing-related features. Finally, the weight W is element-wise multiplied with the input feature map F', that is, $F^* = W \bigotimes F'$, to obtain the final output feature map F^*. After a subsequent convolutional operation, the final output F'' is produced with dimensions $C' \times H \times W$.

E. Loss Function

To ensure stable training and optimized detection, WDHN-Det adopts YOLOv11's standard loss function strategy, combining classification loss, bounding box regression loss, and distribution focal loss (DFL). The total loss is the following equation:

$$L = \lambda_{\text{cls}} L_{\text{cls}} + \lambda_{\text{box}} L_{\text{box}} + \lambda_{\text{dfl}} L_{\text{dfl}} \tag{7}$$

where $\lambda_{\text{cls}} = 0.5$, $\lambda_{\text{box}} = 7.5$, and $\lambda_{\text{dfl}} = 1.5$ are hyperparameter weights following YOLOv11 configuration.

The classification loss L_{cls} employs binary cross-entropy with logits loss (BCEWithLogitsLoss)

$$L_{\text{cls}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log\left(\sigma(p_i)\right) + (1 - y_i)\log\left(1 - \sigma(p_i)\right)\right] \tag{8}$$

where N is the number of positive samples, y_i is the ground-truth label, p_i is the predicted logits, and $\sigma(\cdot)$ is the sigmoid function. This emphasizes hard-to-classify samples, enhancing class discrimination.

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The bounding box regression loss L_{box} utilizes complete IoU loss (CIoU), considering overlap, center distance, and aspect ratio

$$L_{\text{box}} = \frac{1}{N}\sum_{i=1}^{N}\left(1 - \text{CIoU}\left(b_i^p, b_i^g\right)\right) \tag{9}$$

where b_i^p and b_i^g are predicted and ground-truth boxes. CIoU is the following equation:

$$\text{CIoU} = \text{IoU} - \frac{\rho^2\left(c^p, c^g\right)}{d^2} - \alpha v. \tag{10}$$

Here, $\rho^2(c^p, c^g)$ is the squared distance between centers, d is the diagonal length, and v measures aspect ratio consistency. This effectively handles multiscale targets and complex backgrounds.

The DFL L_{dfl} models fine-grained box distributions via Softmax across four directions

$$L_{\text{dfl}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{k=0}^{K-1} y_{i,k}\log\left(\hat{y}_{i,k}\right) \tag{11}$$

where $K = 16$ discretization bins, $y_{i,k}$ is the ground-truth distribution, and $\hat{y}_{i,k}$ is the predicted probability. This enhances boundary uncertainty modeling, improving small target robustness.

Positive samples are assigned via a task-aligned assigner with dynamic weighting to adapt to heterogeneous optical and SAR features. Experiments validate significant improvements in convergence and accuracy on RSOD and SSDD datasets.

IV. Experiment and Discussion

This section describes the experimental setup, including dataset configuration, training settings, while comprehensively evaluating the proposed WDHN-Det method. Several factors were considered: training configurations, dataset details, and comparisons with different approaches. Ablation studies further validate and optimize the key components of the technique. Visualization results intuitively demonstrate the advantages and capabilities of our method in addressing cross-modal feature alignment, multiscale feature degradation, and complex background interference in multisource remote sensing object detection.

A. Experimental Settings

1) Datasets: We conducted extensive validation experiments on two commonly used remote sensing image datasets, RSOD and SSDD, to evaluate the proposed WDHN-Det model. In addition, we compared WDHN-Det with existing state-of-the-art (SOTA) detection methods. The results demonstrate that WDHN-Det outperforms the competing approaches in terms of detection performance.

1) The RSOD dataset [35] was developed by RSIA-LIESMARS at Wuhan University. It comprises aerial images annotated with axis-aligned bounding boxes for four object categories (airplanes, playgrounds, overpasses, and oil tanks), enabling precise localization in remote-sensing tasks. An 8:2 training/validation split was adopted.
2) The SAR ship detection dataset (SSDD) dataset [36] is a widely used open benchmark for ship detection in SAR imagery. It includes 1,160 images from TerraSAR-X, RadarSat-2, and Sentinel-1 with spatial resolutions ranging from 1 to 15 m across HH, VV, HV, and VH polarizations. Ships are divided into large, medium, and small sizes, with the majority being small. Following strict rules, images whose file numbers end in "1" or "9" form the test set; the remainder forms the training set, yielding an approximately 8:2 split consistent with prior works.

2) Implementation Details: The experiments were conducted on Ubuntu 20.04 LTS, equipped with an Intel[1] Core[2] i9-9900K CPU at 3.60 GHz ×16 and an NVIDIA GeForce RTX 2080Ti GPU with 11 GB memory. The deep learning framework was PyTorch 2.5.1, running with CUDA 12.2. To ensure fair and consistent evaluation, all comparative and ablation studies were performed without using any pretrained weights. Training was carried out with batch sizes of 8 or 16 over 300 epochs, employing an SGD optimizer with a momentum of 0.937, an initial learning rate of 0.01, and weight decay set to 0.0005.

B. Critical Design Choices and Validation

1) Frequency-Domain Method Comparison Analysis: To validate wavelet-based frequency-domain pooling superiority, we compared FFT, DCT, and Haar wavelets. Energy conservation testing (Table II) shows the Haar wavelet achieves 100% energy preservation, satisfying Parseval's theorem, while DCT and FFT exhibit 0.76% and 0.83% losses. Fig. 6 visualizes characteristics: FFT shows center-concentrated spectrum, DCT energy concentrates in upper-left region, Haar decomposes into four subbands (LL: 98.8%); reconstruction reveals FFT/DCT blurring while Haar perfectly matches the original.

Edge preservation testing further validates the advantages. Haar achieves SSIM 0.8824, PSNR 28.24 dB, and 64.10% edge preservation, outperforming competitors. Fig. 7 demonstrates: Haar exhibits the highest clarity in building edges; error maps show minimal uniform deviation for Haar versus stripe-like errors (DCT) and diffuse errors (FFT); edge detection confirms Haar preserves 64.1% original edges versus DCT (54.86%) and FFT (62.10%).

Haar's advantages stem from multiresolution time–frequency joint analysis. Unlike FFT's global decomposition and DCT's block transform, wavelets achieve precise frequency localization via scale dilation and spatial translation. Low-frequency subbands ensure structure propagation while directional high-frequency subbands enable edge extraction. Perfect reconstruction guarantees avoid FFT's Gibbs oscillations and DCT's blocking artifacts, making Haar the optimal choice for frequency-domain pooling.

2) DyHPSA Noise Suppression Capability Analysis: To validate the noise suppression capability of the DyHPSA module, we selected representative samples from the validation set and comparatively analyzed the feature map statistical characteristics at the 10th backbone layer between the baseline model and DyHPSA-enhanced model. As shown in

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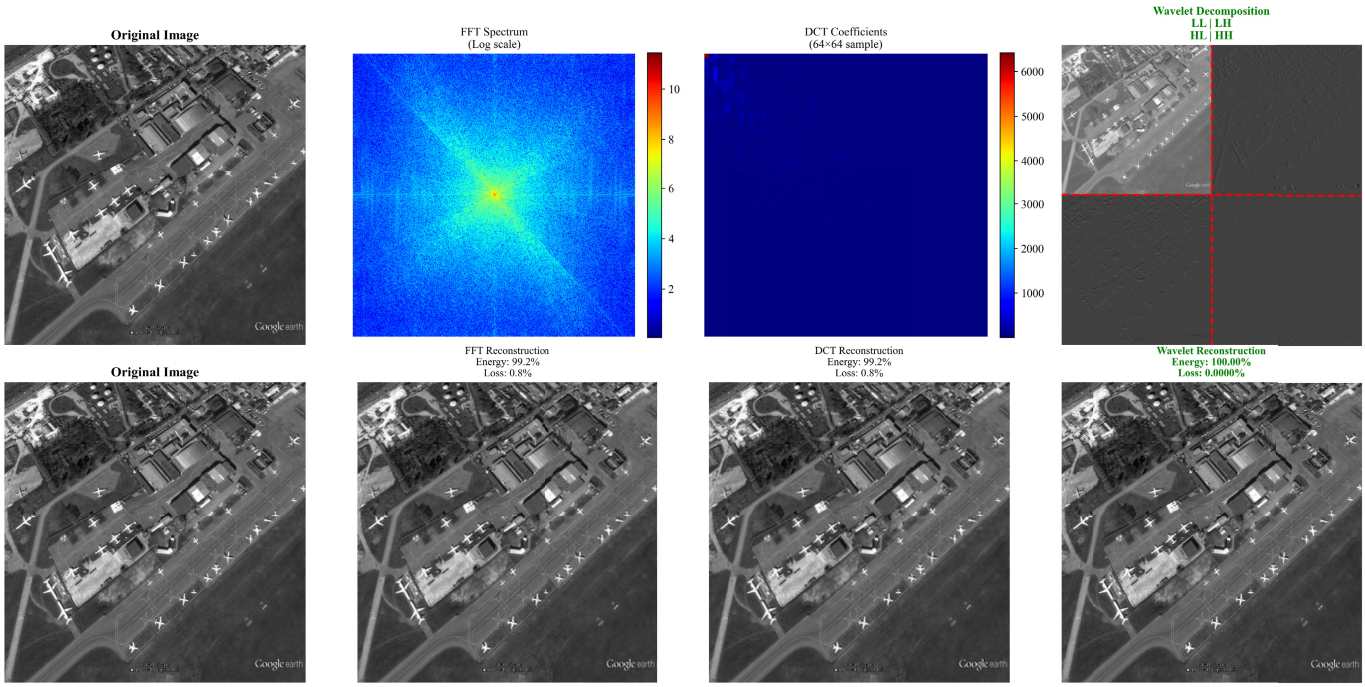


Fig. 6. Information preservation comparison.

TABLE II
COMPREHENSIVE PERFORMANCE COMPARISON OF FREQUENCY-DOMAIN METHODS

Method	Energy Retention	Energy Loss	SSIM 42 *↑	PSNR (dB) ↑	Edge Retention ↑
Haar Wavelet	100.00%	0.00%	0.8824	28.24	64.10%
Discrete Cosine Transform (DCT)	99.24%	0.76%	0.8682	27.81	54.86%
Fourier Transform (FFT)	99.17%	0.83%	0.8609	27.70	62.10%

TABLE III
COMPARISON OF FEATURE STATISTICAL CHARACTERISTICS IN DYHPSA

Configuration	Dataset	Mean	Std	Max	Outlier %
Baseline	RSOD	0.149	0.048	0.348	1.25%
+ DyHPSA	RSOD	0.213	0.045	0.341	0.00%
Baseline	SSDD	0.190	0.070	0.445	1.00%
+ DyHPSA	SSDD	0.174	0.068	0.349	0.00%

Figs. 8 and 9, DyHPSA achieves significant noise suppression effects across both datasets: on the RSOD optical dataset, feature standard deviation decreases from 0.048 to 0.045 (5.5% reduction), outlier ratio is completely eliminated from 1.25% to 0%, and the coefficient of variation in feature distribution decreases by 33.9%; on the SSDD SAR dataset, maximum amplitude significantly decreases from 0.445 to 0.349 (21.7% reduction), standard deviation decreases from 0.070 to 0.068, and outliers are similarly completely eliminated. Feature amplitude distribution histograms clearly demonstrate that DyHPSA transforms the originally dispersed feature distribution (blue) into a more compact Gaussian-like distribution (red), with difference heatmaps further revealing that noise suppression primarily occurs in high-variance regions while preserving critical structural information such as target boundaries.

Quantitative results (Table III) demonstrate that DyHPSA automatically learns parameter configurations adapted to different modalities through data-driven approaches: the RSOD model's adaptive parameters are 0.44 and 0.62 (mean 0.53 ± 0.09), emphasizing variance reduction and distribution concentration to address illumination variations and sensor noise; the SSDD model's α values are 0.22 and 0.66 (mean 0.44 ± 0.22), with larger parameter differences reflecting more aggressive dynamic range adjustment strategies for SAR speckle noise. The achievement of 100% outlier elimination across both datasets serves as a critical indicator, demonstrating that DyHPSA effectively constrains extreme-valued features, providing more stable and discriminative feature representations for subsequent detection tasks.

C. Experimental Results and Comparisons

We conducted comparative experiments using standard evaluation metrics in object detection to comprehensively assess WDHN-Det's performance. These metrics include: mAP@0.5 for measuring detection accuracy, Params for evaluating model complexity, and giga floating-point operations (GFLOPs) for quantifying computational cost. These metrics align with the standard evaluation frameworks proposed in [18], [37] on the RSOD dataset and [4] on the SSDD dataset, ensuring the comparability and reliability of the experimental results.

Table IV presents a comprehensive comparison between WDHN-Det and various SOTA detectors on the RSOD dataset.

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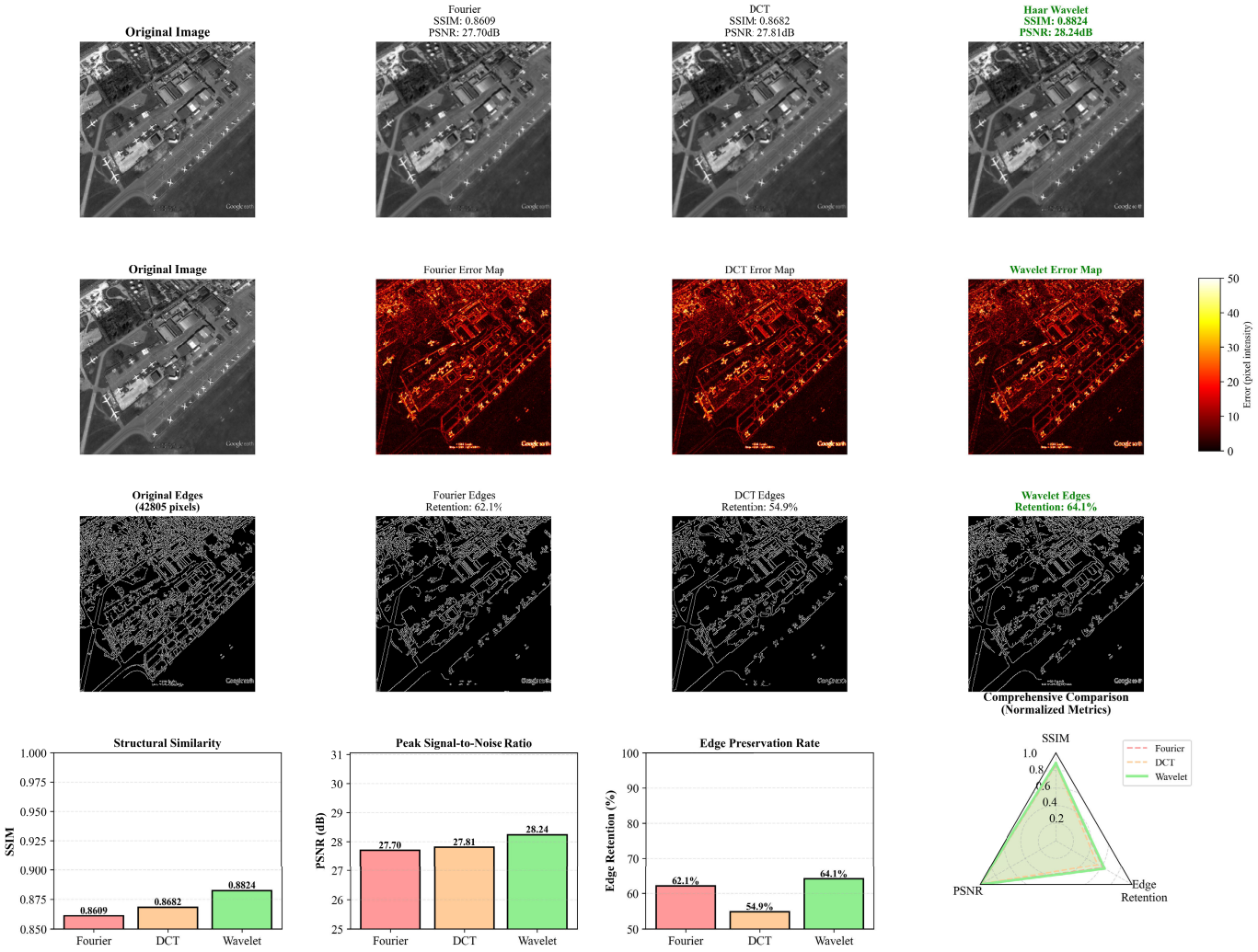


Fig. 7. Reconstruction quality and edge preservation comparison.

TABLE IV

COMPARISON OF MAP PARAMS AND GFLOPS FOR DIFFERENT MODELS ON RSOD

Method	mAP.5	Params	GFLOPs	Airplane	Oil tank	overpass	playground
Faster R-CNN [38]	84.47	41.7	69.29	70.84	90.19	78.74	98.09
Cascade R-CNN [39]	91.30	63.5	140.20	94.20	96.10	83.20	99.00
YOLOv5 [40]	89.50	46.5	116	88.90	95.20	80.30	98.70
YOLOv8 [41]	90.10	43.7	118.9	90.50	95.70	81.50	98.90
FCOS [42]	86.20	32.0	133.3	76.30	92.80	79.00	98.20
DETR [43]	87.80	41	187	75.50	93.50	79.80	98.50
RT-DETR [44]	91.10	41	136	93.20	96.00	82.90	99.00
DINO [45]	92.50	47	210	94.00	96.50	84.50	99.30
Deformable-DETR [46]	90.70	40.5	173	92.50	95.80	83.50	**99.50**
Enhanced RT-DETR [18]	93.8	45.5	150	95.50	**98.50**	81.90	**99.50**
MSNet [37]	95.62	29.81	98.15	**96.35**	97.31	90.86	97.97
YOLOv11s [47]	93.36	9.41	21.3	95.27	97.27	81.68	99.22
WDHN-Det(ours)	**95.86**	**8.99**	**18.2**	94.79	98.07	**91.08**	**99.50**

WDHN-Det achieves the highest mAP@0.5 of 95.86%, surpassing the second-best MSNet by 0.24% while demonstrating superior efficiency (Params: 8.99 M, FLOPs: 18.2 G)—69.8% fewer parameters and 81.5% lower computational cost than MSNet. Compared to Transformer-based models DINO and Deformable-DETR, WDHN-Det improves accuracy by 3.36% and 5.16%, respectively, with over 80% parameter reduction and 89% FLOPs decrease. Against single-stage detectors

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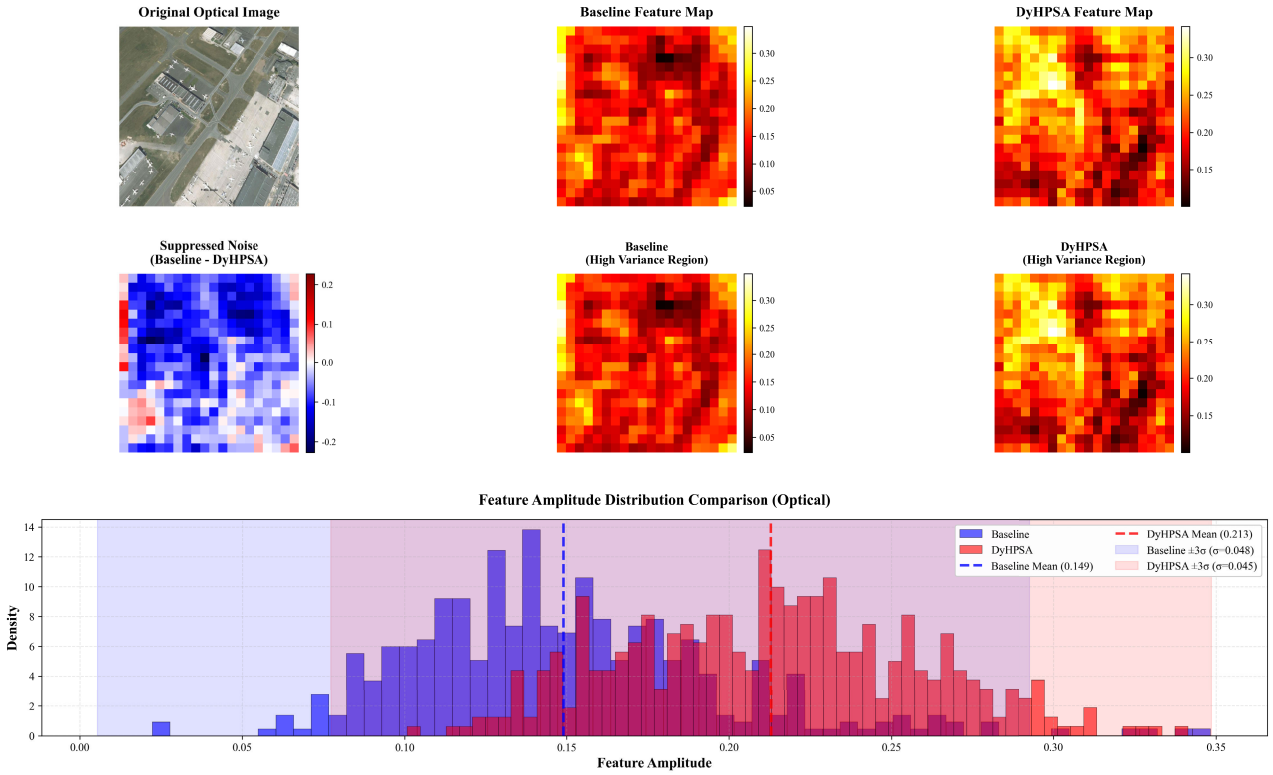


Fig. 8. Optical image noise suppression analysis.

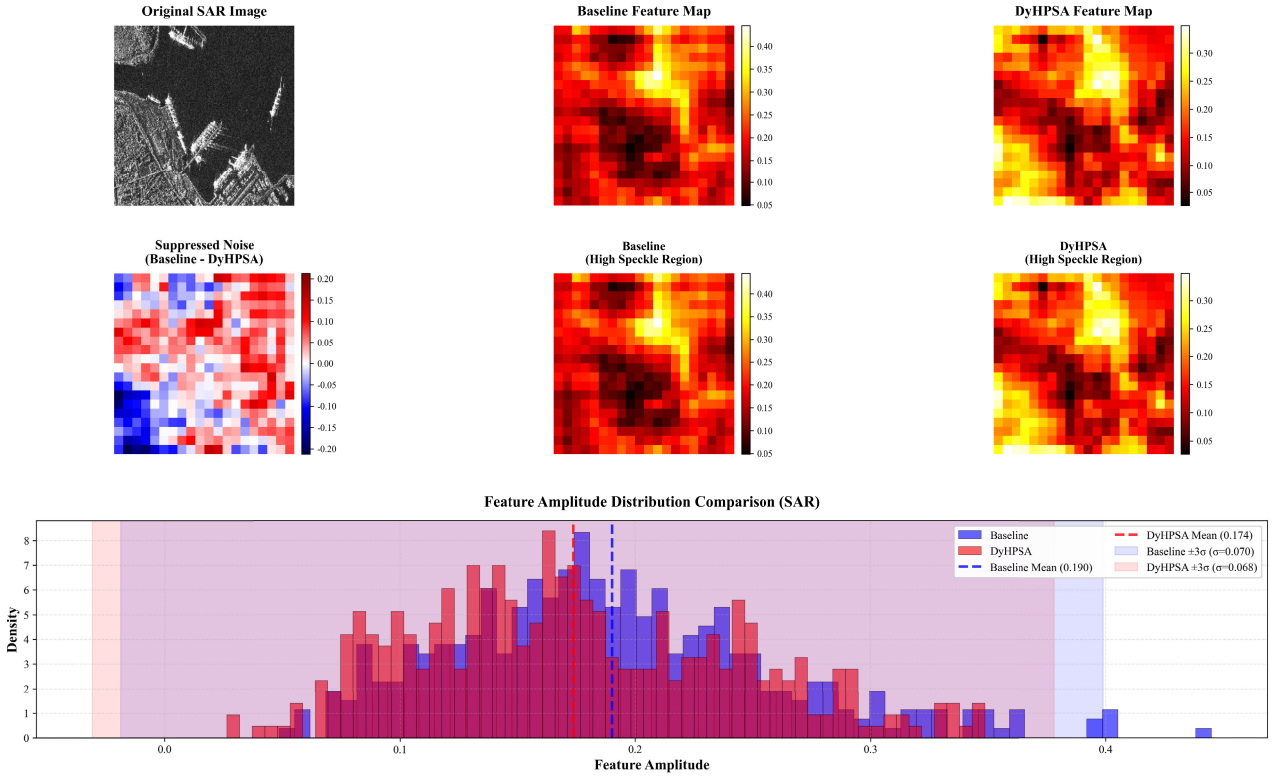


Fig. 9. SAR image noise suppression analysis.

YOLOv8 and YOLOv11s, it gains 2.50%–5.76% in mAP@0.5 with further efficiency optimization. Per-class AP on challenging categories like overpass significantly outperforms baselines (9.4% improvement over YOLOv11s),

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TABLE V
COMPARISON OF MAP, PARAMS, AVERAGE INFERENCE TIME FOR DIFFERENT MODELS ON SSDD

Method	mAP.5	Params	FLOPs
Faster R-CNN [38]	96.4	41.12	91.41
Cascade R-CNN [39]	96.8	69.17	119.05
FBR-Net [48]	94.1	32.5	——
CornerNet [49]	94.7	201.04	707.75
CenterNet [50]	95.1	14.1	20.40
FCOS [42]	95.3	60.97	78.63
AFSar [51]	97.7	——	8.66
LPEDet [52]	97.4	5.68	18.38
CenterNet++ [53]	95.1	——	——
YOLOv3 [54]	96.2	61.52	77.54
YOLOv5n [40]	98.1	1.76	4.1
YOLOX [55]	96.2	12.45	41.78
HRLE-SARDet [4]	98.4	**1.09**	**2.5**
SARNas [5]	98.5	2.31	2.6
SARATR-X [6]	97.3	——	——
YOLOv11s [47]	98.3	9.41	21.3
WDHN-Det (ours)	**98.7**	8.99	18.2

demonstrating WDHN-Det's balanced superiority in accuracy and efficiency for multisource remote sensing detection.

Table V demonstrates WDHN-Det's performance on the SSDD dataset. WDHN-Det achieves 98.7% mAP@0.5, surpassing the second-best model by 0.2%–0.6% and significantly outperforming traditional methods such as Faster R-CNN and YOLOv3 (2.3%–2.5% improvement). In terms of model complexity, its parameter count (8.99 M) and FLOPs (18.2 G) are substantially lower than two-stage detectors while remaining comparable to lightweight models. The inference time (10.72 ms) outperforms Faster R-CNN and YOLOv3, achieving an excellent balance between accuracy and efficiency. Compared to lightweight models LPEDet and AFSar, WDHN-Det gains 1.0%–1.3% in mAP, demonstrating robust advantages in handling speckle noise and multiscale targets in SAR ship detection.

Results from Tables V and VI reveal that using identical network architecture and training strategies, WDHN-Det surpasses domain-specific methods on both optical dataset RSOD (95.86% mAP@0.5) and SAR dataset SSDD (98.73% mAP@0.5). This cross-modal consistency indicates that the three core modules successfully incorporate modality-invariant design principles, enabling the same feature extraction and fusion mechanisms to adapt to heterogeneous data sources.

Detection comparisons in Figs. 10 and 11 clearly illustrate the performance improvements of WDHN-Det over the baseline. In Fig. 10, the top row displays RSOD images, and the bottom row shows detection results, where red circles mark false positives and green circles indicate regions with marked accuracy gains. WDHN-Det effectively reduces false detections in the airplane category, precisely localizes small aircraft, detects more oil tanks with fewer false alarms, and achieves more uniform detections along overpasses. For playground targets, false positives in open areas are substantially decreased, demonstrating improved robustness under complex backgrounds. Fig. 11 presents SSDD SAR images and detection comparisons; WDHN-Det significantly lowers false and missed detections, particularly in dense port areas, yielding more accurate and balanced results. These visual comparisons further validate the complementary roles of CSWFF, DyHPSA, and CM-AFGPF modules in jointly enhancing detection accuracy and robustness by suppressing false positives and reducing missed detections in multisource remote sensing object detection.

In summary, WDHN-Det achieves outstanding performance on object detection tasks over the RSOD and SSDD remote sensing datasets. The CSWFF module employs Haar wavelet transforms to preserve high-frequency details and multiscale features, enhancing sensitivity to small targets. DyHPSA improves target–background discrimination by adaptively modulating feature magnitudes via dynamic hyperbolic normalization combined with pyramid attention. CM-AFGPF further boosts detection accuracy and robustness through channel–spatial attention-based fusion of cross-modal features, effectively capturing fine-grained information in complex environments.

D. Module Ablation Experiments

1) Core Module Effectiveness Validation: Tables VI and VII present the ablation study results of the WDHN-Det framework on the RSOD and SSDD datasets, evaluating the individual contributions and synergistic effects of the CSWFF, DyHPSA, and CM-AFGPF modules. Using YOLOv11s as the baseline, the experiments progressively incorporate each module to analyze their impacts on detection performance, computational complexity, and efficiency.

Analysis shows that introducing DyHPSA alone elevates RSOD's mAP@0.5 to 93.56% and $F1$-score to 91.06%, demonstrating its dynamic attention mechanism optimizes feature robustness; on SSDD, mAP@0.95 rises to 74.96%, indicating enhanced discrimination in complex backgrounds. Adding CM-AFGPF alone achieves RSOD's mAP@0.5 of 94.97% and $F1$-score of 92.34%, with SSDD's precision reaching 96.84%, validating its role in cross-modal feature fusion. Incorporating CSWFF significantly optimizes efficiency, boosting RSOD's mAP@0.5 to 95.24% while reducing GFLOPs to 18.2, parameters to 7.94 M, and increasing FPS to 253.95; on SSDD, mAP@0.5 holds at 98.25% with substantial reductions in GFLOPs and parameters, underscoring its benefits in multiscale feature extraction. Combining CM-AFGPF and DyHPSA yields synergistic effects, raising RSOD's mAP@0.5 to 95.39% and SSDD's to 98.39%; full integration of all modules attains RSOD's mAP@0.5 of 95.86% and $F1$-score of 92.91%, with SSDD's mAP@0.5 reaching 98.73%, markedly improving overall performance and efficiency. In summary, the ablation experiments validate the WDHN-Det framework's efficacy. CSWFF lowers computational complexity and bolsters multiscale feature extraction, DyHPSA enhances feature discriminability, and CM-AFGPF improves cross-modal fusion. Their synergy achieves comprehensive optimization of precision, recall, and efficiency on the RSOD and SSDD datasets, demonstrating robust adaptability in complex remote sensing scenarios.

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TABLE VI
ABLATION STUDY OF WDHN-DET ON THE RSOD DATASET

Method	P	R	mAP.5	mAP.95	F1	GFlOPs	Params	FPS	Model_size
baseline	88.40	90.93	93.36	64.73	89.63	21.3	9.41	268.48	18.3
+DyHPSA	91.91	90.33	93.56	65.07	91.06	21.3	9.45	255.07	18.3
+CM-AFGPF	95.83	89.29	94.97	65.64	92.34	21.3	10.46	262.30	20.3
+CSWFF	94.96	91.82	95.24	**65.84**	**93.34**	**18.2**	**7.94**	**253.95**	**15.5**
+CM-AFGPF +DyHPSA	94.51	**91.86**	95.39	65.18	93.14	21.3	10.46	232.51	20.3
+DyHPSA+CSWFF+CM-AFGPF	**95.08**	91.00	**95.86**	**64.92**	**92.91**	**18.2**	**8.99**	**218.71**	**17.5**

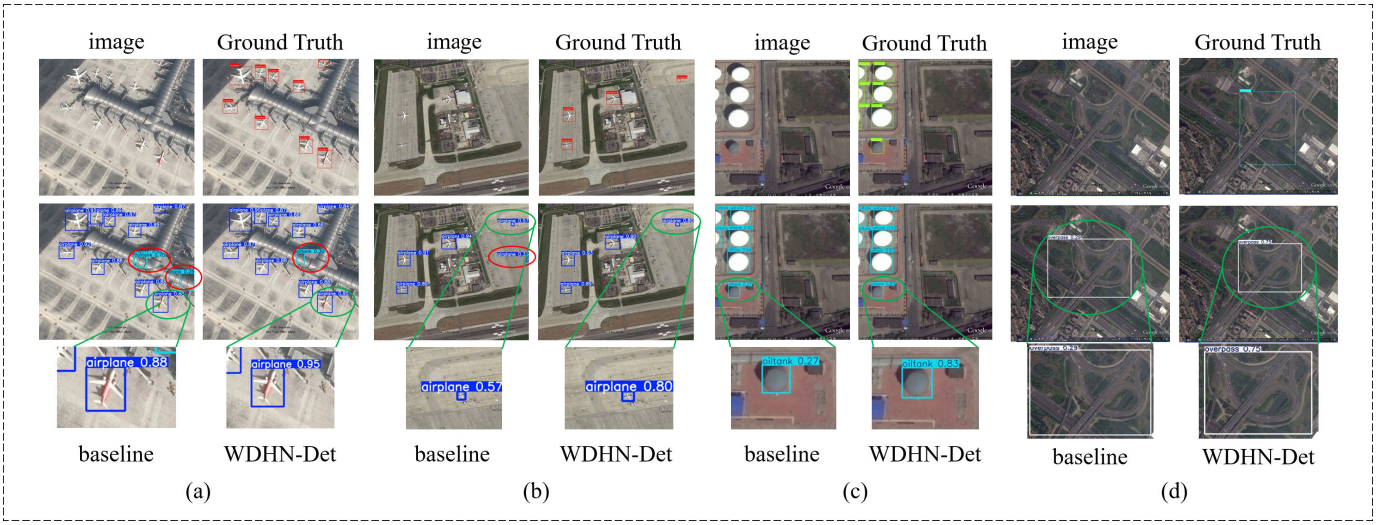


Fig. 10. Comparison of detection results between WDHN-Det and the baseline on the RSOD dataset. Red circles indicate false positives, while green circles highlight areas with significant accuracy improvement.

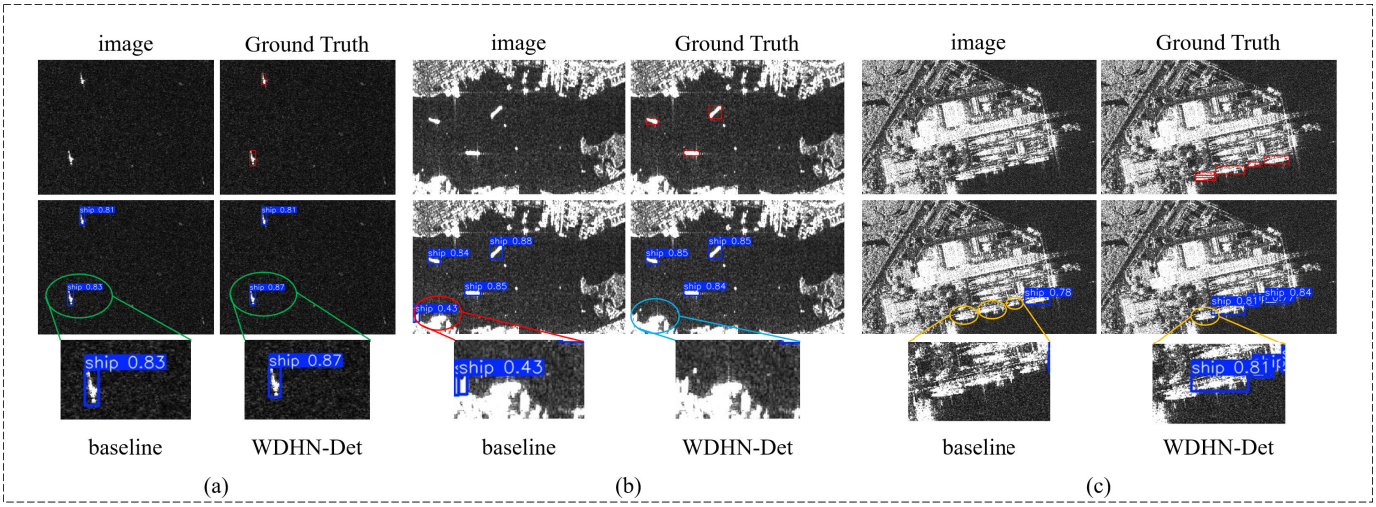


Fig. 11. Comparison of detection results between WDHN-Det and the baseline on the SSDD dataset. Red circles indicate false positives, blue circles mark regions correctly avoided by WDHN-Det, green circles highlight areas with significant accuracy improvement, and orange circles denote missed detections.

2) Normalization Strategy Ablation Analysis: The choice of normalization layer significantly impacts the feature representation capability of deep networks. In the design of the DyHPSA module, we adopted dynamic Tanh normalization instead of traditional BN or LN. To validate the rationale of this design choice, this section systematically compares the performance of six mainstream normalization strategies within the DyHPSA module. Table VIII presents a detailed performance comparison of different normalization strategies on the RSOD and SSDD datasets.

On the RSOD dataset, DyHPSA (using dynamic Tanh normalization) achieved the highest mAP@0.5 of 95.86% and mAP@0.95 of 64.92%, outperforming the PSA baseline 93.24% mAP@0.5, RMSNormPSA 94.21%, LayerNormPSA

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TABLE VII
ABLATION STUDY OF WDHN-DET ON THE SSDD DATASET

Method	P	R	mAP.5	mAP.95	F1	GFLOPs	Params	FPS	Model_size
baseline	95.65	95.43	98.32	74.04	**96.14**	21.3	9.41	**312.15**	18.3
+DyHPSA	96.44	94.36	98.08	**74.96**	95.39	21.3	9.41	290.56	18.3
+CM-AFGPF	**96.84**	95.39	98.25	73.44	96.11	21,3	10.46	288.87	20.3
+CSWFF	95.78	**95.52**	98.25	73.30	95.65	**18.2**	**7.93**	265.11	**15.5**
+CM-AFGPF +DyHPSA	96.65	95.02	98.39	74.21	95.83	21.3	10.46	279.60	20.3
+DyHPSA+CSWFF+CM-AFGPF	96.31	95.42	**98.73**	72.80	**95.87**	**18.2**	**8.99**	**261.79**	**17.5**

TABLE VIII
ABLATION STUDY ON DYTANH NORMALIZATION STRATEGY ON RSOD AND SSDD

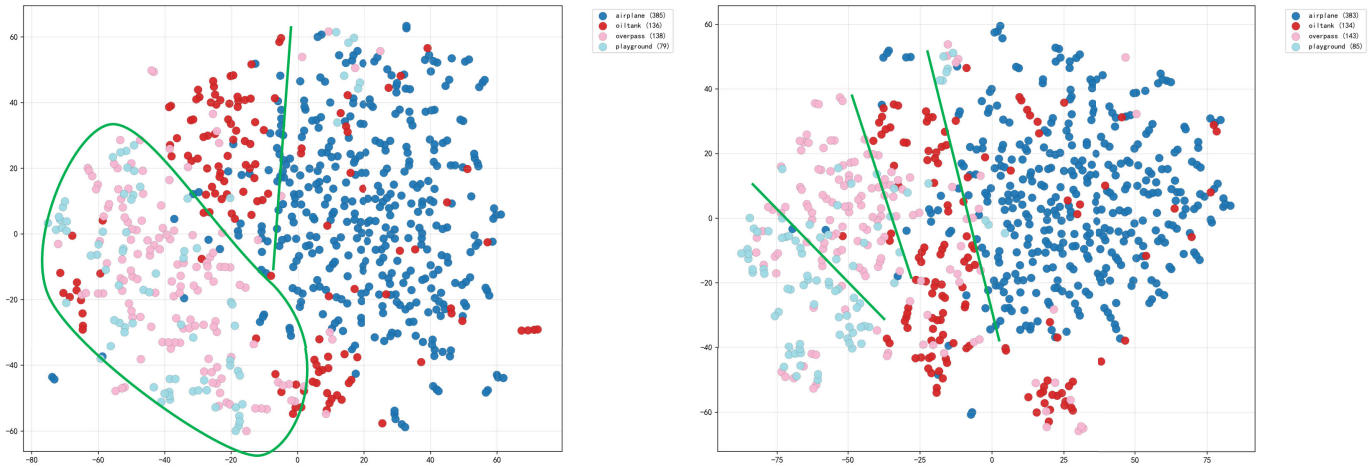
Method	dataset	P	R	mAP.5	mAP.95	F1	GFLOPs	Params	FPS	Model_size
A+B+PSA	RSOD	91.13	90.09	93.24	63.25	90.61	18.2	8.99	241.05	17.5
A+B+RMSNormPSA	RSOD	91.36	**91.75**	94.21	63.78	91.51	18.2	8.99	**255.34**	17.5
A+B+LayerNormPSA	RSOD	93.84	91.29	95.40	64.75	92.52	18.2	8.99	215.61	17.5
A+B+BatchNormPSA	RSOD	93.82	89.68	94.37	63.53	91.68	18.2	8.99	229.49	17.5
A+B+GroupNormPSA	RSOD	94.36	90.86	95.27	64.33	92.50	18.2	8.99	221.71	17.5
A+B+InstanceNormPSA	RSOD	90.95	89.88	92.35	63.31v	90.04	18.2	8.99	228.09	17.5
A+B+DyHPSA	RSOD	**95.08**	91.00	**95.86**	**64.92**	**92.91**	**18.2**	**8.99**	218.71	**17.5**
A+B+PSA	SSDD	94.68	95.42	98.22	72.98	95.05	18.2	8.99	246.08	17.5
A+B+RMSNormPSA	SSDD	96.21	95.42	98.03	73.73	95.82	18.2	8.99	253.65	17.5
A+B+LayerNormPSA	SSDD	95.86	**96.34**	98.69	73.52	96.10	18.2	8.99	239.46	17.5
A+B+BatchNormPSA	SSDD	**97.55**	94.66	98.03	73.48	96.08	18.2	8.99	249.34	17.5
A+B+GroupNormPSA	SSDD	96.96	95.24	97.78	73.35	96.09	18.2	8.99	236.63	17.5
A+B+InstanceNormPSA	SSDD	97.38	95.22	98.14	**74.02**	**96.29**	18.2	8.99	245.55	17.5
A+B+DyHPSA	SSDD	96.31	95.42	**98.73**	72.80	95.87	**18.2**	**8.99**	**261.79**	**17.5**

TABLE IX
COMPARISON OF CM-AFGPF WITH OTHER ATTENTION MECHANISMS ON RSOD AND SSDD

Method	dataset	P	R	F1	mAP.5	mAP.95	GFLOPs	Params	FPS	Model_size
Baseline	RSOD	88.40	90.93	89.63	93.36	64.73	21.3	9.41	**263.93**	18.3
A+C+SPPF	RSOD	88.52	90.27	89.34	91.61	63.80	18.2	**7.94**	257.51	**15.5**
A+C+SPPF_SE	RSOD	94.11	89.22	91.50	94.68	63.79	18.2	8.07	248.41	15.8
A+C+SPPF_EMA	RSOD	93.14	91.57	92.33	94.85	63.20	19.1	8.10	211.26	15.8
A+C+SPPF_SimAM	RSOD	**95.89**	92.25	**94.00**	**96.21**	64.21	18.2	**7.94**	218.41	**15.5**
A+C+SPPF_LSKA	RSOD	89.69	**93.83**	91.65	94.18	63.92	19.0	9.00	224.01	17.6
A+C+SPPF_CBAM	RSOD	92.92	93.19	93.04	94.64	64.18	18.2	8.07	212.73	15.8
A+C+SPPF_BAM	RSOD	90.82	92.71	91.74	94.65	63.89	18.3	8.26	239.79	16.1
A+C+CM-AFGPF	RSOD	95.08	91.00	92.91	95.86	**64.92**	**18.2**	8.99	**218.71**	17.5
Baseline	SSDD	95.65	96.63	96.14	98.32	74.05	21.3	9.41	**312.15**	18.3
A+C+SPPF	SSDD	94.91	95.57	95.24	97.95	73.61	18.2	**7.94**	257.51	**15.5**
A+C+SPPF_SE	SSDD	97.12	93.77	95.42	98.10	73.30	18.2	8.07	248.30	15.8
A+C+SPPF_EMA	SSDD	**97.18**	94.32	95.73	98.20	73.82	19.1	8.10	256.25	15.8
A+C+SPPF_SimAM	SSDD	96.47	95.24	95.85	98.42	73.99	18.2	**7.94**	257.54	**15.5**
A+C+SPPF_LSKA	SSDD	96.82	94.71	95.75	98.01	73.72	19.0	9.00	247.16	17.6
A+C+SPPF_CBAM	SSDD	96.53	96.89	**96.71**	**98.84**	**74.22**	18.2	8.07	246.30	15.8
A+C+SPPF_BAM	SSDD	95.34	**97.48**	96.40	98.60	73.52	18.3	8.26	256.74	16.1
A+C+CM-AFGPF	SSDD	96.31	95.42	95.87	98.73	72.80	**18.2**	8.99	**261.79**	**17.5**

95.40%, BatchNormPSA 94.37%, GroupNormPSA 95.27%, and InstanceNormPSA 92.35%. It also excelled in precision 95.08% and $F1$-score 92.91% while maintaining consistent efficiency metrics (18.2 GFLOPs, 8.99 M Params). Similarly, on the SSDD dataset, DyHPSA attained the highest mAP@0.5 of 98.73%, surpassing most variants and demonstrating enhanced robustness to speckle noise without compromising FPS (261.79). These results underscore the efficacy of dynamic

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(a) (b)

Fig. 12. t-SNE visualization results on the RSOD dataset. (a) Baseline t-SNE on RSOD. (b) WDHN-Det t-SNE on RSOD.

Tanh normalization in balancing noise suppression and feature discriminability, making it the optimal choice for our remote sensing framework requirements.

3) Attention Method Ablation Analysis: We compared the standard SPPF and its variants integrated with six mainstream attention mechanisms (SE, EMA, SimAM, LSKA, CBAM, and BAM) to validate the effectiveness of the CM-AFGPF design. As shown in Table IX, CM-AFGPF achieves superior cross-modal performance. On RSOD, it reaches 95.86% mAP@0.5, a 4.25% gain over SPPF. Although SimAM attains 96.21%, its accuracy drops on SSDD. On SSDD, CM-AFGPF achieves 98.73% mAP@0.5, slightly below CBAM and SimAM, but its average mAP@0.5 across both datasets (97.30%) surpasses all counterparts, demonstrating the best cross-modal stability. Single attention mechanisms often excel in specific domains but struggle with heterogeneous optical–SAR distributions. CM-AFGPF integrates fine-grained pyramid fusion, dual (channel–spatial) attention, and deformable pooling, ensuring robustness on both RSOD's complex geometries (P = 95.08% versus CBAM's 92.92%) and SSDD's speckle-noise scenarios (mAP@0.95 = 72.80%). With only 8.99 M parameters (+ 13.2%) and 18.2 GFLOPs, it significantly improves cross-modal detection (RSOD + 4.25%, SSDD + 0.78%) over SPPF, validating its fine-grained design for multisource remote sensing. The standard SPPF's lowest accuracy further confirms the necessity of attention mechanisms for cross-modal fusion.

E. Visualization Result

1) Feature Distribution Visualization: Fig. 12 visualizes the t-SNE feature distributions on the RSOD dataset, providing an intuitive comparison between the baseline and WDHN-Det models. In the baseline (left), significant feature entanglement is observed—particularly between overpass (pink) and playground (cyan) classes, which overlap densely within the green elliptical region. This indicates poorly defined interclass boundaries and limited discriminative capacity, often leading to false positives. In contrast, WDHN-Det (right) yields more compact and well-separated clusters for airplane (blue), oiltank (red), overpass (pink), and playground (cyan), with markedly reduced confusion between visually similar categories such as overpass and playground. From a cross-modal feature learning perspective, the pronounced class separation capability is significant: t-SNE projections reveal the intrinsic structure of the high-dimensional feature space learned by WDHN-Det. Its ability to establish discriminative feature boundaries on the RSOD optical dataset demonstrates that the extracted features transcend superficial textures, capturing the geometric and semantic essence of targets. This semantic-level representation is crucial for cross-modal alignment. Despite stark visual differences between ship targets in the SSDD SAR dataset (characterized by scattering intensity) and aircraft or overpasses in RSOD (defined by RGB textures), they share high-level semantic commonalities, such as distinct boundaries and specific shapes. Through CSWFF's frequency-domain decomposition, DyHPSA's dynamic calibration, and CM-AFGPF's fine-grained fusion, WDHN-Det effectively captures these shared semantic features. This accounts for the framework's superior performance across heterogeneous optical and SAR data, as it learns modality-invariant discriminative semantic representations rather than modality-specific superficial features.

2) Heatmap Comparative Analysis: Additionally, heatmap visualizations on the RSOD and SSDD datasets (Figs. 13 and 14) intuitively demonstrate the model's attention distribution over target regions. On RSOD, the baseline model's heatmaps show dispersed and weakly correlated activations (bluish areas) around overpass and playground targets, whereas WDHN-Det exhibits more concentrated and intense red activations, especially on complex geometric targets. This enhanced focus correlates with the mAP@0.5 improvement from 93.36% to 95.86%, reflecting superior localization precision. On SSDD, baseline heatmaps reveal notable blue interference in dense ship clusters, indicating low target correlation, while WDHN-Det achieves more uniform and stronger activations, reducing

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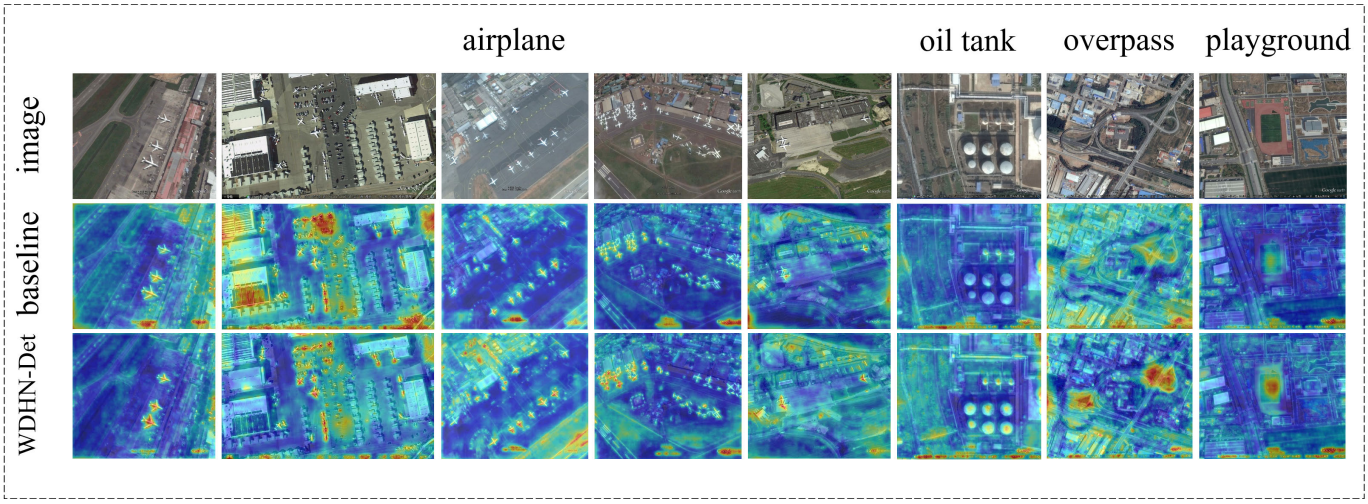


Fig. 13. Heatmap comparison of the original image, baseline, and WDHN-Det on the RSOD dataset. Red regions indicate correlation with the target category, with deeper red signifying higher correlation; blue regions indicate lower correlation.

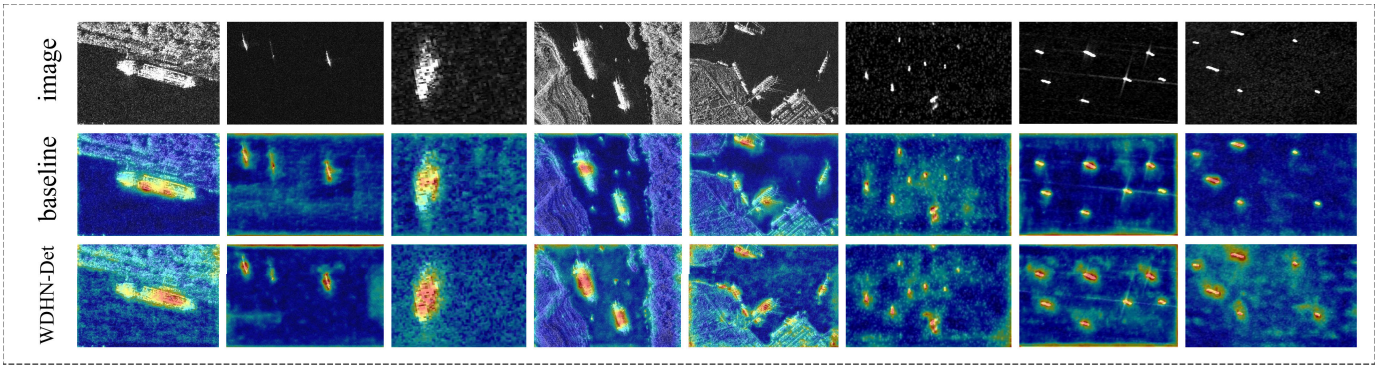


Fig. 14. Heatmap comparison of the original image, baseline, and WDHN-Det on the SSDD dataset. Red regions indicate correlation with the target category, with deeper red signifying higher correlation; blue regions indicate lower correlation.

false positives. This aligns with the mAP@0.5 increase from 98.32% to 98.73%, confirming improved robustness. These findings further validate the complementary roles of CSWFF, DyHPSA, and CM-AFGPF modules in enhancing detection accuracy.

V. Conclusion

We propose WDHN-Det, a wavelet-coordinated dynamic hyperbolic normalization framework for multisource remote sensing object detection. The core innovation lies in the synergistic design of three modules—CSWFF, DyHPSA, and cross-modality adaptive fine-grained pyramid fusion—effectively addressing key technical challenges in multisource remote sensing detection. This framework tackles heterogeneity between optical and SAR imaging mechanisms, insufficient multiscale feature representation, and complex background interference. Extensive experiments on the RSOD and SSDD datasets validate WDHN-Det's superiority in detection accuracy and model efficiency, outperforming multiple SOTA methods. Results demonstrate that the framework effectively enhances the accuracy and stability of multisource remote sensing object detection. Despite significant performance gains, WDHN-Det has limitations: it is currently designed for optical and SAR modalities, requiring validation for generalization to additional modalities (e.g., infrared and hyperspectral); while computationally optimized, real-time deployment on edge devices may face challenges. Future work will focus on further enhancing model generalization and optimizing module designs to adapt to more complex remote sensing scenarios.

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